STEM Fellowship Inter-University Big Data Challenge 2022

Health Insurance Grouping

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Abstract

Innovation and advancements in medicine and technology have given birth to a new myriad of medical procedures, each targeting specific demographics. As a result, it has become more difficult to gauge a customer's medical expenses. The inability to accurately estimate a customer's expected medical bill is a problem for health insurance companies as they often end up mischarging them due to the variation between each individual's expenses. Here, a customer's demographic is used to predict whether the customer is within a high-cost or low-cost group. This solution can be made into a more precise model using unsupervised machine learning by employing clustering techniques that would find implicit patterns that are impossible to find manually. The findings were that the most consistently high-cost customers were those with a high BMI and who smoked. These findings also tell us that other factors, such as gender, are not deemed to be major factors behind a customer's medical expenses. Health insurance companies can use these findings to revise and create more accurate pricing models.

Keywords
Health Economics, Medical Insurance, Unsupervised Machine Learning, Gaussian Mixture Model

1 Introduction

Health is an important factor in leading a long life. However, the state of an individual's health is unpredictable and fluctuates. The information collected for risk assessment is self-reported [1] and taken from previous medical records and questionnaires filled out during the beginning of a contract [2]. Health insurance companies conduct risk assessments to mitigate the risk of falling ill or getting injured in exchange for a premium [3].

The US has much higher insurance charges compared to other developed regions around the world such as Canada and Europe [4, 5]. According to the Kaiser Family Foundation, the average insurance premiums per year in the United States for an individual is $7,739 and $22,221 for a family [6]. Although there are many factors relating to these high costs, including high healthcare ad-

ministration costs, excessive spending due to greed, and higher utilization of medical expenditure on technology [4], this study will focus on how an individual's behaviour and demographic affect their insurance charges.

There are a few related works that have been conducted to predict the cost of health insurance premiums. These works include studies on identifying the most appropriate insurance plan for an individual and improving risk management [7], implementing and comparing the accuracy of different regression models to predict insurance costs [8], the role of pharmacotherapy on healthcare costs [9], using deep neural networks architecture to predict healthcare costs from health insurance claim records [10], and supervised learning methods for predicting healthcare costs [11]. In comparison, this study aims to determine and group the factors that contribute to higher insurance costs using unsupervised machine learning.

The objective of this study is to determine and group the demographic and behavioural factors that lead to the highest charges as insurance companies assess the risk of customers.

2 Materials & Methods

2.1 Sourcing the Dataset

The dataset is originally from the "Machine Learning with R" book, by Brett Lantz. This dataset is available on GitHub and Kaggle.

2.2 Data Pre-processing

The dataset was in a comma-separated value (CSV) format which was imported into a Jupyter notebook using the pandas library. The non-numerical columns were converted into multiple binary columns using OneHotEncoder [12]. Finally, all the columns had to be normalized with the help of StandardScalar [13].

2.3 Model Selection and Evaluation

All the models made use of unsupervised learning algorithms. Evaluation and selection of clusters are subjective to each individual, as there are no reliable criteria to judge. The data was processed through K-Means, Density-Based Spacial Clustering of Application with Noise (DBSCAN), Hierarchical DBSCAN (HDBSCAN) and Gaussian Mixture Models (GMM) models.

All the models were visualised on scatter plots, to better judge the quality of clusters. GMM was selected as the best clustering model, followed closely by HDBSCAN. K-Means and DBSCAN performed poorly.

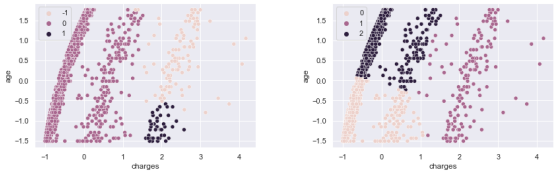


Figure 1: Poor Clusters formed by K-Means and DBSCAN

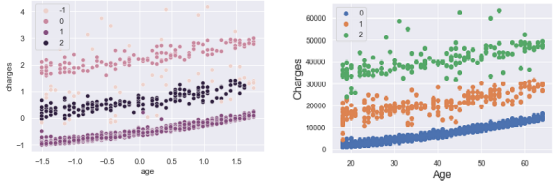


Figure 2:
Better Clusters formed by HDBSCAN and GMM

2.4 Assessing feature importance

Feature importance was calculated using a logistic regression model. After fitting

the data, the absolute value of the coefficient for each scaled feature was directly used to judge its importance. As the features are scaled, the coefficient, or weighting, gives a direct quantity of the importance of a variable.

3 Results

3.1 Which factors affected charges the most?

The most interesting feature in relation to charges was age. When plotting the distribution of charges by age, the charges split into three very distinct groups.

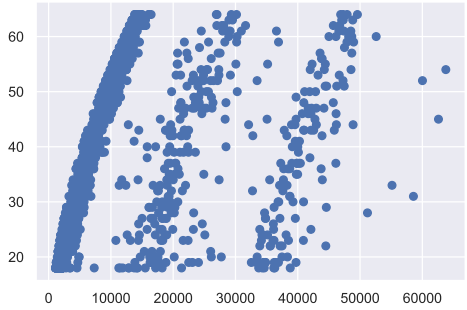


Figure 3: Scatterplot of Charges vs Age

As the age increases, the charge per group increases, and the older people from the previous group intersect with younger people from the next more expensive group. Because of this, without factoring in age it is easy to miss the clear separation between these groups.

Clustering models can be used to obtain labels for the different groups, and quantify the differences between them. The Gaussian Mixture Model technique was used to create the clusters, as is shown in the following graph:

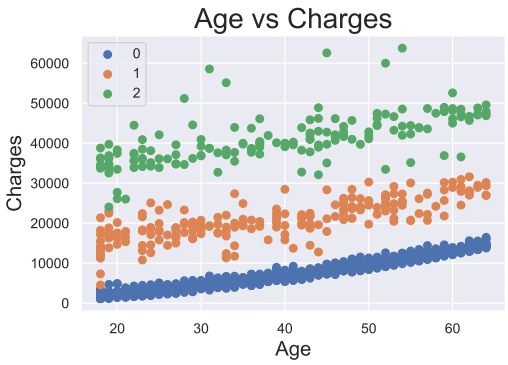


Figure 4: Clusters visualised by the GMM

The Lowest charge group is band A, the middle is band B and the highest charge is band C.

3.2 Cluster analysis

The differences in charges between the bands are significant, as shown in the figure below.

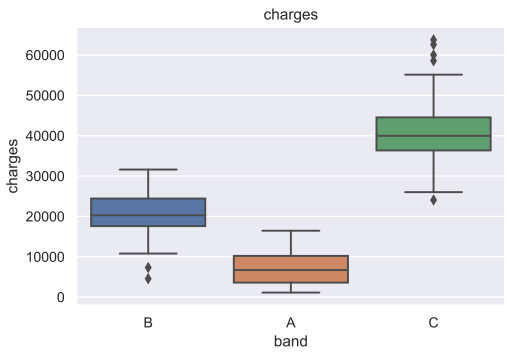


Figure 5: Box plot of Charges

Band A has a median charge of approximately $7,500, while B has a median of $20,00 and C $40,000.

The first feature investigated is BMI. Below is a box plot of BMI for all bands.

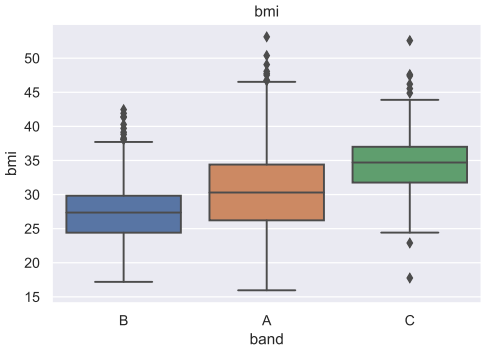


Figure 6: Box plot of BMIs

It is interesting to see that the BMI of band B is less than band A, but the following figure of BMI by bands split by smoking gives a more complete picture;

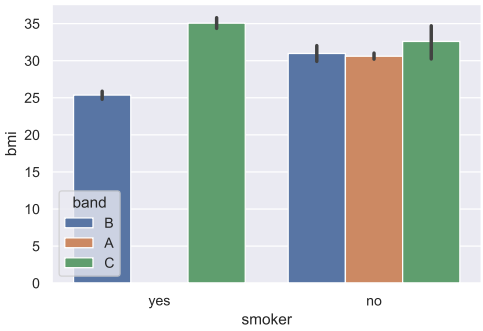


Figure 7: BMI bar graph separated by whether they smoke

The average BMI increases with the bands in the non-smoking group, while in the smoking group band B has a significantly lower average BMI.

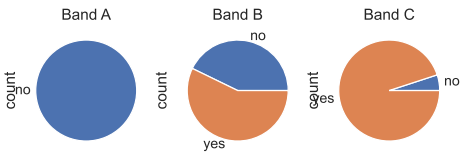


Figure 8: Smokers distribution in each band

The figure above shows the ratio of smokers to non-smokers by the band. It can be observed that the higher bands have a much higher proportion of smokers than non-smokers.

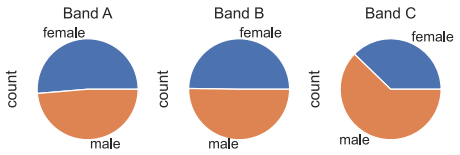


Figure 9: Gender distribution in each band

This figure shows the proportion of gender in each band, with the first two being close to equal, and band C having a larger proportion of males.

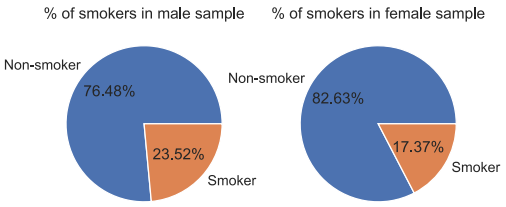


Figure 10: Gender proportions by smoking

Males seem to smoke more than females, as shown by the graph above.

Finally, below is a table of the most important features, found by creating a logistic regression model, and the values are the coefficients of the features.

	Attribute	Importance	Positive
0	Smoker	2.758618	True
1	Non-smoker	2.757531	False
2	BMI	0.598217	False
3	Northeast	0.222024	False
4	Southwest	0.159069	True
5	No_of_Children	0.104026	False
6	Northwest	0.048638	True
7	Male	0.024245	True
8	Female	0.023158	False
9	Age	0.021578	True
10	Southeast	0.015404	True

Figure 11: Regression coefficients

4 Discussion

4.1 Clustering

Several techniques were used to create the clusters for the different bands. Initially, K-Means was used, but it performed poorly, as shown in figure 1, as it is reliant on the data being in spherical groups, while the groups here are offset columns. The next technique used was DBSCAN, which also did not perform optimally as it is a density-based method, but the bands are of disproportionate densities. To solve this, a technique called HDBSCAN was made [14]. This is less impacted by differing densities, and it performs much better than the previous two techniques. GMM clustering was also used, shown in Figure 2, and this was decided to be the most optimal of the two techniques, as it separated well between the groups and wasn't as impacted by noise as HDBSCAN was.

4.2 Highest contributing factors

Looking at the factors which affect the bands, smoking seems to have an extremely significant effect, as shown in figure 10. Band A has no smokers, while band B has what appears to be two-thirds and band C consists of over 90% smokers.

The BMI also combines with smoking to produce a very noticeable difference. Figure 7 shows that the BMI for smokers is much lower than non-smokers for band B, while the BMI for both in band C is large, suggesting that only smokers with lower than average BMIs are in band D, and a combination of smoking and high BMI makes you much more likely to be in band C.

This is compounded by the feature importance shown in Figure 11. Smoking has a feature importance, or weighting, of negative 2.76 while BMI has a weighting of negative 0.6. A negative weighting will increase a cluster's chance of going into a higher band, so those two features having high negative weighting show that they combine to increase the band, and thereby increase the charges.

Another interesting feature is gender. Figure 9 shows that bands A and B have close to equal proportions of males and females, but band C has a much higher proportion of males. It appears to be close to a two-to-one proportion of males to females. Looking deeper into the data set, it appears that the reason behind that is because a higher amount of males smoke relative to females, as shown in figure 10.

4.3 An in-depth analysis of the lowest charges group

Band A is the group with the lowest mean charges by a significant amount. It also contains the majority of the patients from the data, nearly three-quarters. That means the majority of customers medical insurance companies deal with belong to this band. It has the largest sample size which also means that any trends uncovered can be stated with much more confidence than that of the other two bands. It's for all these reasons above that this paper will only look at band A.

For example, when looking at how the number of children affects the average charges, there's a clear positive correlation. At least at the start. The relationship takes a strange dip when the number of children reaches 4 for the data set, and 5 for our band A. It seems counter-intuitive, after all the greater number of children on your insurance should imply higher charges on average.

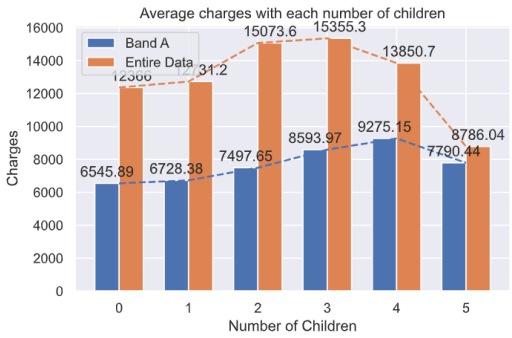


Figure 12: Family sizes in band A

Upon further inspection, the reason for one of the line's dips was uncovered. For the full dataset, it makes no sense for families with 5 children to have the lowest insurance charges out of all sizes. But it turns out that 90% of families with 5 children belong to band A! So it isn't a very fair representation of all three bands of people, with its value essentially being band A's average charge value for families with 5 children. So the next question is how exactly does the family size distribution of band A compare to the rest of the data.

Conclusions

The benefit of determining the primary causes behind the decision of a person's insurance band is that it allows people to quantify how their life choices can impact them financially. Health insurance companies can benefit from this research to provide customers with the appropriate plans that best match their future expected charges based on this model. The model is not complete nor exhaustive by any means. But it can serve as a baseline for more complex models that have more features and larger datasets to train from. Ideally, these future models would be working with industry partners to collect live data and test the model's reliability in real-world use cases. Despite the limited dataset, the study has ultimately shown how a factor like smoking can greatly shift health outcomes.

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